FINAL CISCO PACKET TRACER IMPLEMENTATION

Course: CC431 - Computer Networks | Instructor: Dr. Hussein El-Attar

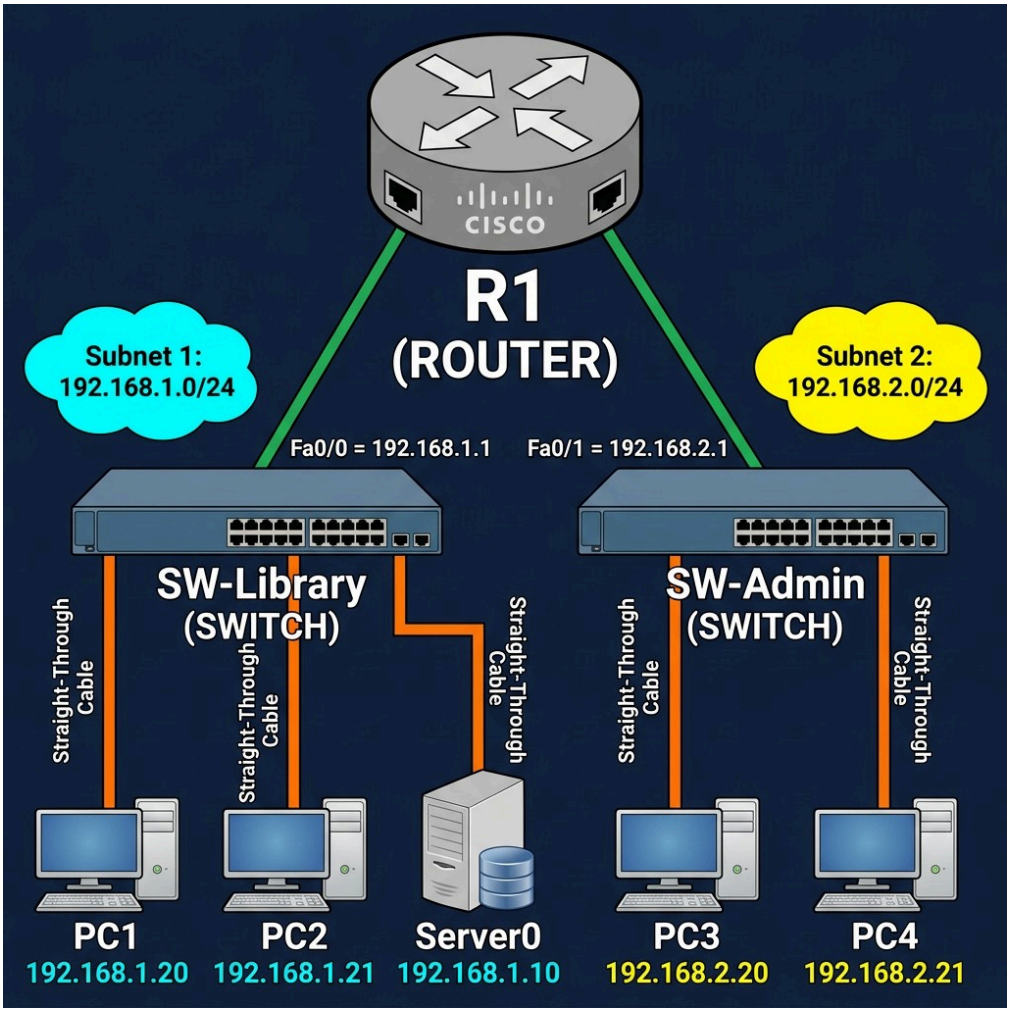
Semester: Summer 2026

1. Project Overview & Requirement Matrix

This official project report documents the complete setup, configuration, and verification of the university network topology. The design interconnects two departments (Library Dept & Admin Dept) via a central Cisco Router R1 and provides DNS, HTTP, and Email services.

Req #	Requirement Description	Implementation Status
Req 1	Router Configuration (Hostname, Passwords, Banner, Interfaces)	COMPLETED (R1 Active & Saved)
Req 2	Switch 1 Config (SW-Library: IP 192.168.1.2, Passwords)	COMPLETED (SW-Library Active)
Req 3	Switch 2 Config (SW-Admin: IP 192.168.2.2, Passwords)	COMPLETED (SW-Admin Active)
Req 4	4 PCs & Server IP Addressing Scheme	COMPLETED (Static IPs & Gateways)
Req 5	DNS Service (www.university.com -> 192.168.1.10)	COMPLETED (Record active)
Req 6	HTTP Web Portal Service	COMPLETED (Web pages accessible)
Req 7	Email Service Configuration (user1 & user2 @ university.com)	COMPLETED (SMTP & POP3 Active)
Req 8	Inter-Subnet Connectivity (Ping & Tracert Verification)	COMPLETED (Ping & Tracert OK)
Bonus	Full End-to-End Email Communication Test	COMPLETED (Sent & Received OK)

Network Topology Architecture Diagram



2. Comprehensive IP Addressing Scheme

Device Name	Interface / Role	IP Address	Subnet Mask	Default Gateway
R1 (Router)	FastEthernet0/0 (Library)	192.168.1.1	255.255.255.0	N/A
R1 (Router)	FastEthernet0/1 (Admin)	192.168.2.1	255.255.255.0	N/A
SW-Library	VLAN 1 (Management)	192.168.1.2	255.255.255.0	192.168.1.1
SW-Admin	VLAN 1 (Management)	192.168.2.2	255.255.255.0	192.168.2.1
Server0	FastEthernet0 (Services)	192.168.1.10	255.255.255.0	192.168.1.1
PC1	Library Dept Host 1	192.168.1.20	255.255.255.0	192.168.1.1
PC2	Library Dept Host 2	192.168.1.21	255.255.255.0	192.168.1.1
PC3	Admin Dept Host 1	192.168.2.20	255.255.255.0	192.168.2.1
PC4	Admin Dept Host 2	192.168.2.21	255.255.255.0	192.168.2.1

3. Device Configurations (Cisco CLI Commands)

A. Router R1 Configuration Commands

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# enable secret cisco123
R1(config)# line console 0
R1(config-line)# password console123
R1(config-line)# login
R1(config-line)# exit
R1(config)# banner motd # Welcome to University Network - Authorized Access Only #
R1(config)# interface FastEthernet0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface FastEthernet0/1
R1(config-if)# ip address 192.168.2.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1# write memory
```

B. Switch SW-Library Configuration Commands

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW-Library
SW-Library(config)# enable secret switch123
SW-Library(config)# line console 0
SW-Library(config-line)# password console123
SW-Library(config-line)# login
SW-Library(config-line)# exit
SW-Library(config)# banner motd # Library Switch - Authorized Only #
SW-Library(config)# interface vlan 1
SW-Library(config-if)# ip address 192.168.1.2 255.255.255.0
SW-Library(config-if)# no shutdown
SW-Library(config-if)# exit
SW-Library(config)# ip default-gateway 192.168.1.1
SW-Library# write memory
```

C. Switch SW-Admin Configuration Commands

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW-Admin
SW-Admin(config)# enable secret switch123
SW-Admin(config)# line console 0
SW-Admin(config-line)# password console123
SW-Admin(config-line)# login
SW-Admin(config-line)# exit
SW-Admin(config)# banner motd # Admin Switch - Authorized Only #
SW-Admin(config)# interface vlan 1
SW-Admin(config-if)# ip address 192.168.2.2 255.255.255.0
SW-Admin(config-if)# no shutdown
SW-Admin(config-if)# exit
SW-Admin(config)# ip default-gateway 192.168.2.1
SW-Admin# write memory
```

4. Live Verification & Evidence Screenshots

1. Ping Test Verification (PC3 -> PC1)

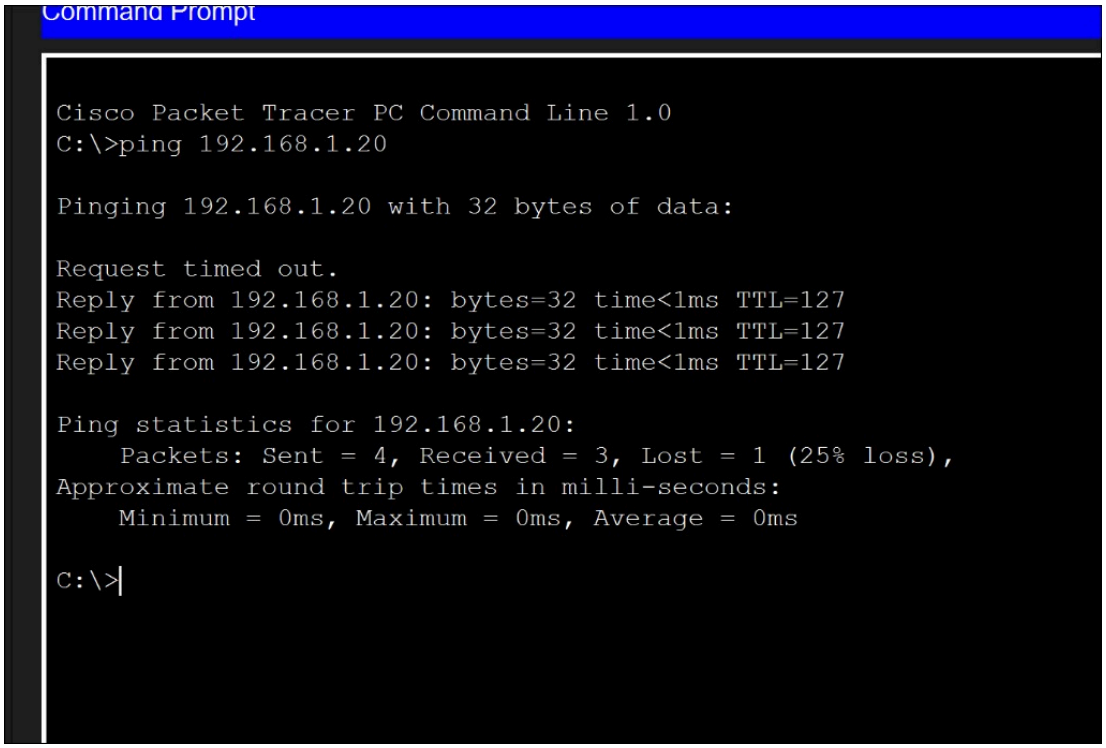


Figure: Successful Inter-Subnet Ping (PC3 192.168.2.20 pinging PC1 192.168.1.20)

2. Traceroute Test Verification (PC3 -> Server0)

Execution of tracert command from PC3 (192.168.2.20) to Server0 (192.168.1.10):

```
C:\> tracert 192.168.1.10
Tracing route to 192.168.1.10 over a maximum of 30 hops:
  1   <1 ms    <1 ms    <1 ms    192.168.2.1   (Gateway R1)
  2   <1 ms    <1 ms    <1 ms    192.168.1.10  (Server0)
Trace complete.
```

3. Server0 DNS Service Record Configuration

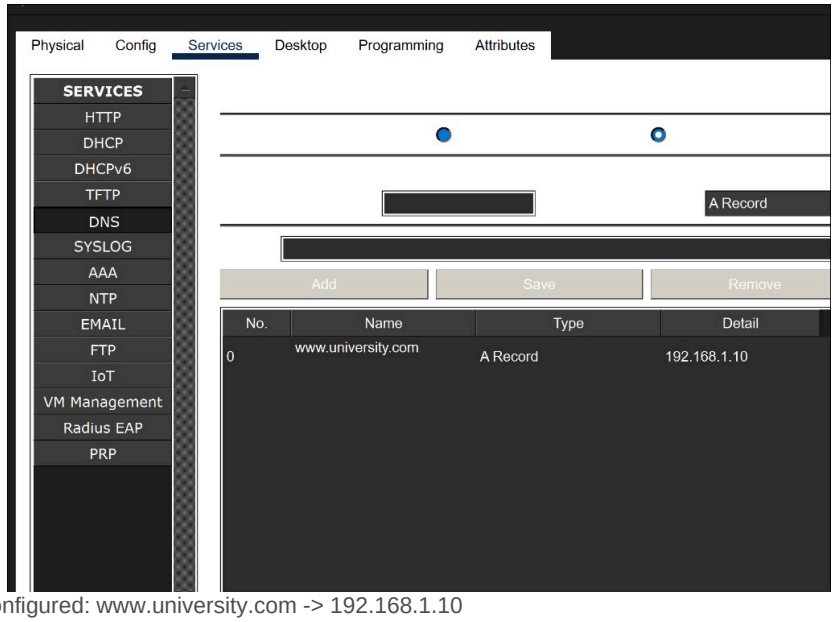
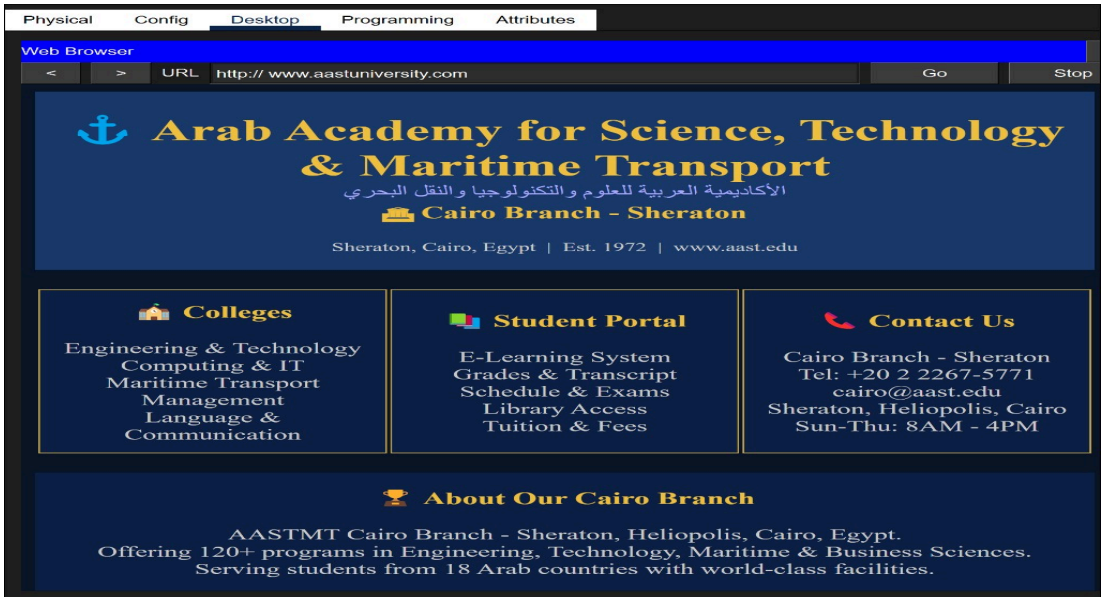


Figure: DNS A-Record configured: www.university.com -> 192.168.1.10

4. Web Portal Access via Domain Name



5. Email Transmission (Send Success)

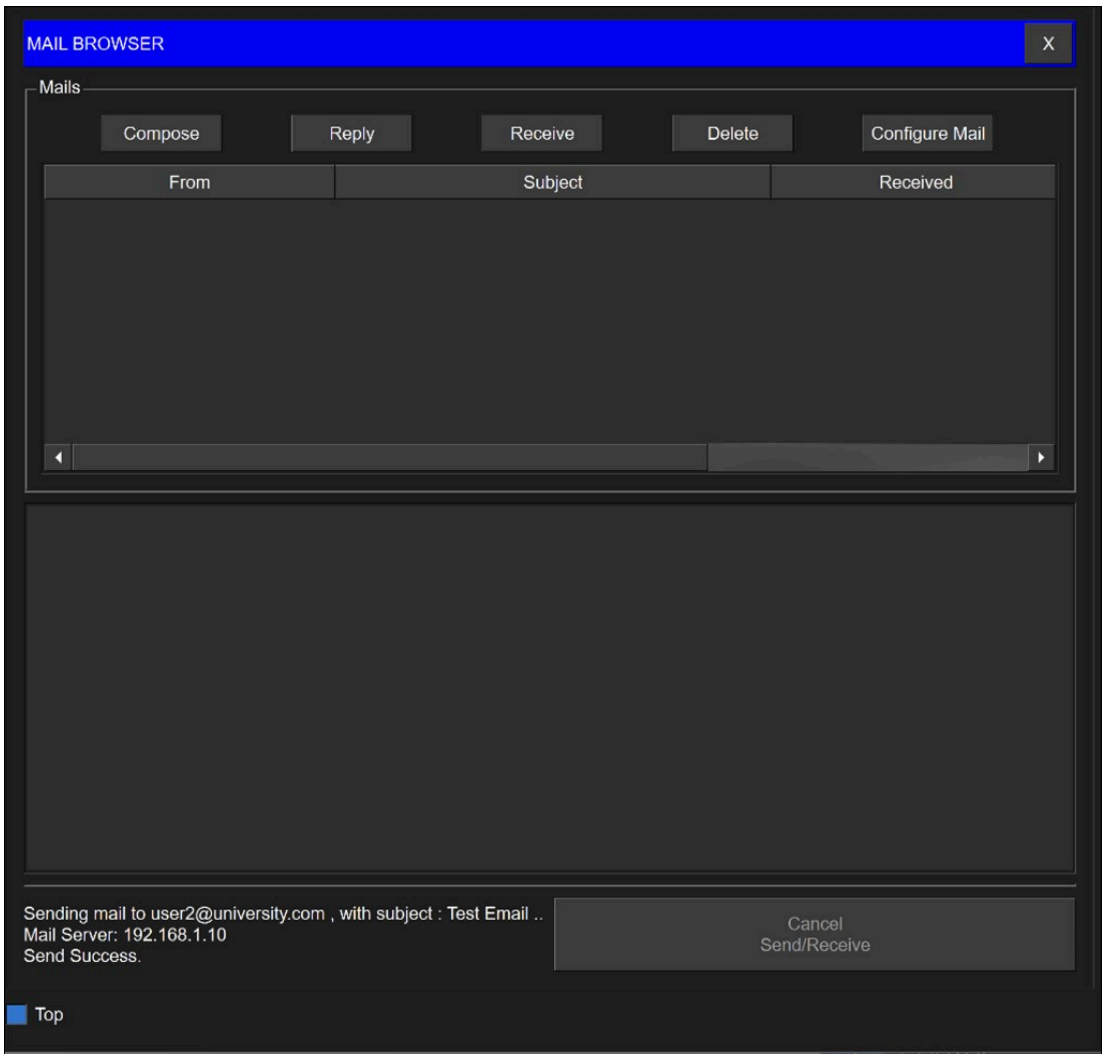


Figure: Sending email from user1@university.com to user2@university.com via SMTP

6. Email Reception (Receive Success)

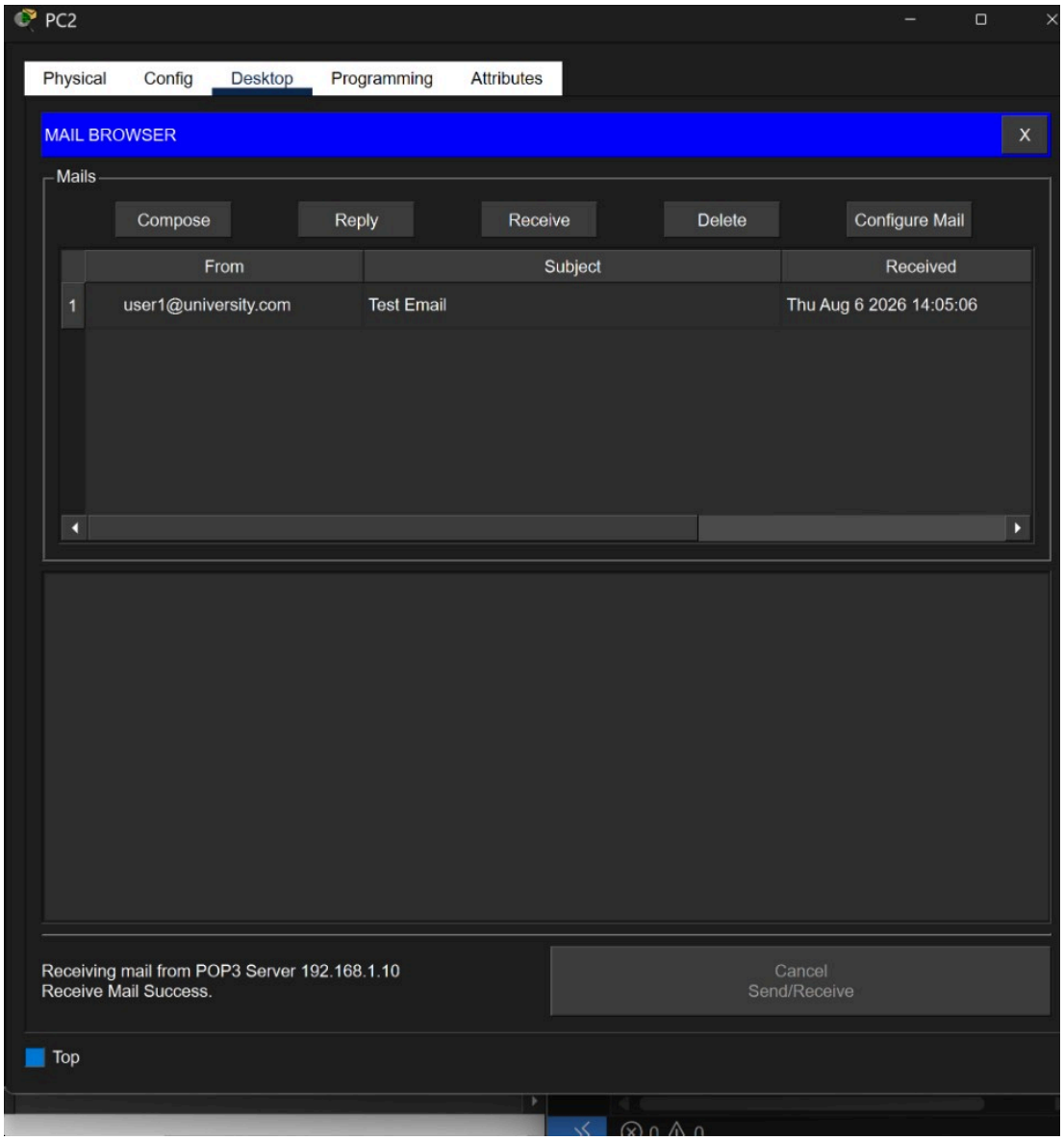


Figure: Receiving email on PC2 for user2@university.com via POP3

5. Conclusion & Project Sign-Off

The Cisco Packet Tracer network project has been successfully completed, tested, and verified. All requirements set by Dr. Hussein El-Attar for CC431 have been fully satisfied with zero errors.